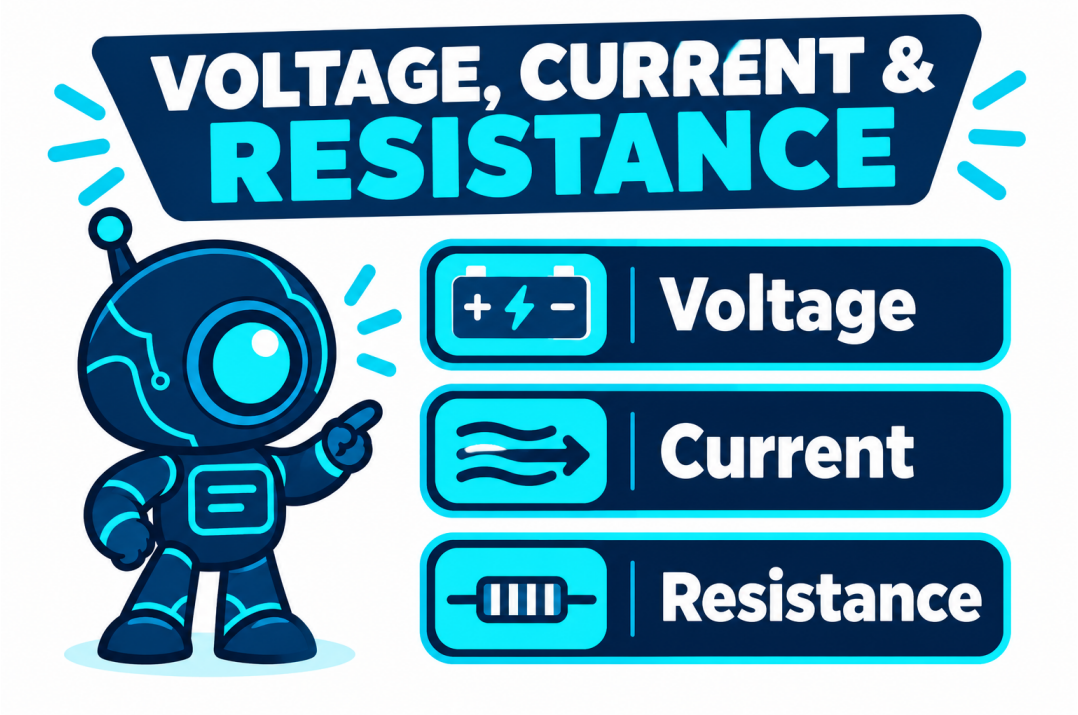
AI MECH LAB

Mechatronics Course — Beginner Track



Module 2, Topic 2.1

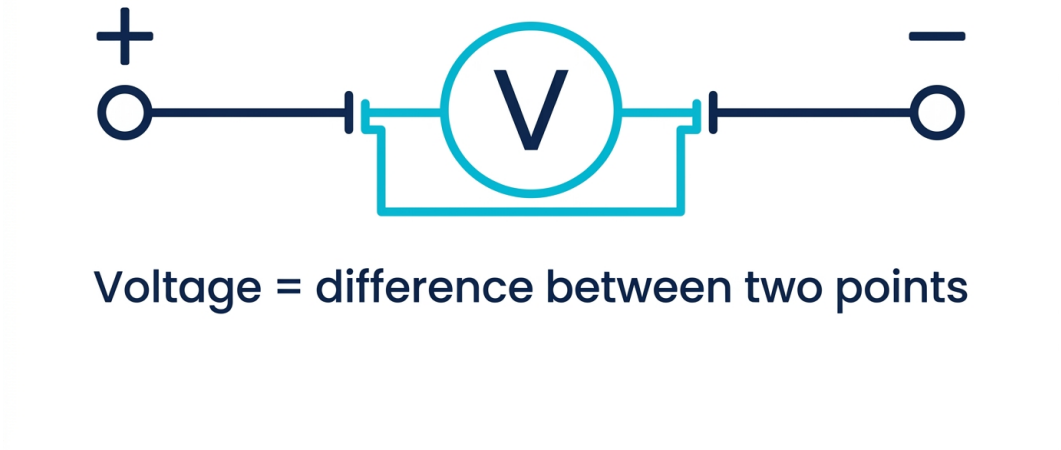
Voltage, Current, and Resistance

Lesson

You've covered the four pillars and how they fit into a real design process. Now you start building the electrical pillar hands-on, beginning with the three quantities every circuit is built from: voltage, current, and resistance. Everything you'll do with sensors and motors later in this course depends on understanding how these three interact.

Voltage — the push

Voltage is the electrical "push" that makes current want to flow. Think of it like water pressure in a pipe: higher pressure pushes more water through, and higher voltage pushes more electrons through a circuit. Voltage is measured in volts (V), and it always exists as a difference between two points — you talk about the voltage across a component, or the voltage between two terminals, never just "the voltage" of a single point in isolation.

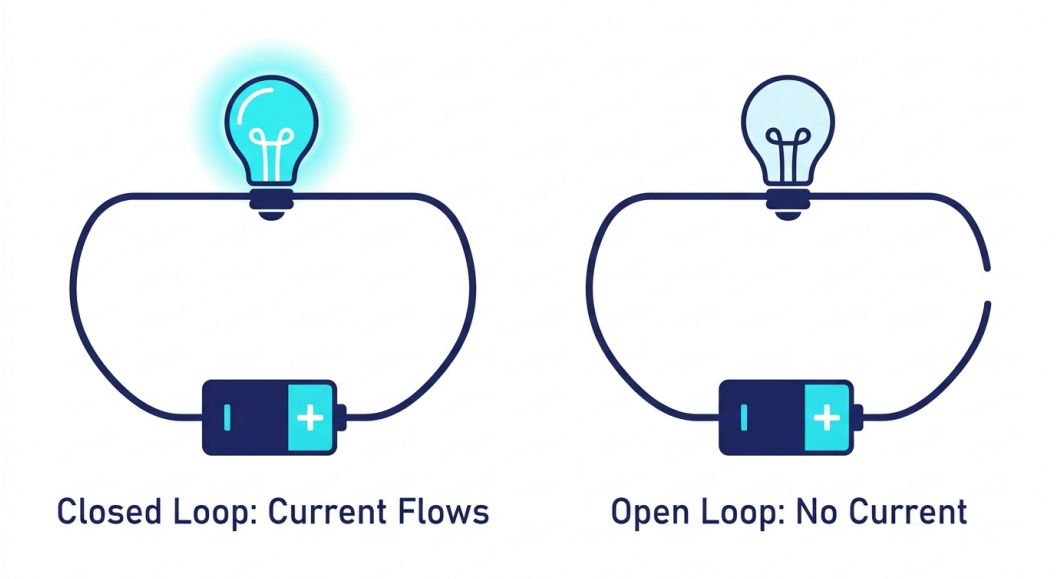


Voltage is always measured as a difference between two points.

A standard AA battery supplies about 1.5V. A USB port typically supplies 5V. These numbers tell you how much electrical push is available to drive current through whatever you connect.

Current — the flow

Current is the actual rate of flow of electric charge through a circuit — continuing the water analogy, if voltage is the pressure, current is the amount of water actually moving through the pipe per second. Current is measured in amperes, usually shortened to amps or A. A small LED might draw around 20 milliamps (0.02A) to light up. A power tool motor might draw several amps under load.

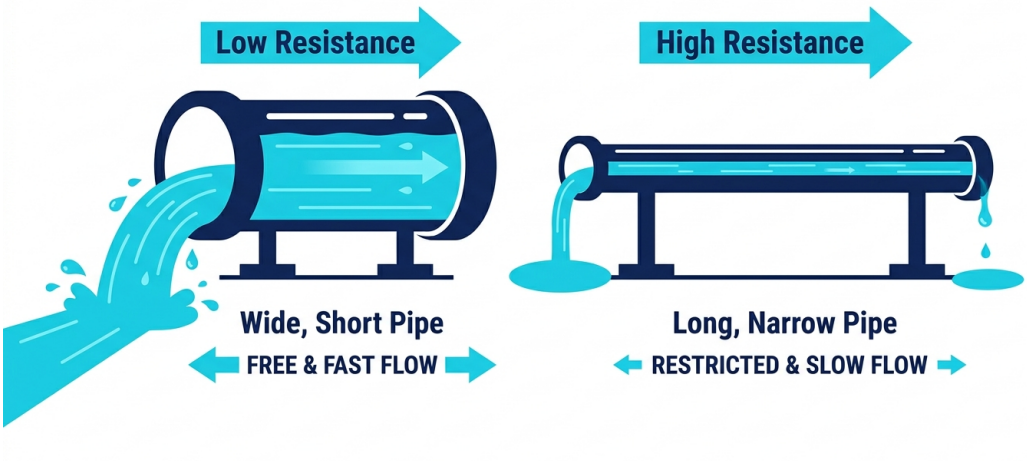


Current only flows through a complete, closed loop.

Current only flows when there's a complete path — a closed loop — for it to travel around. Break the loop anywhere, even with a tiny gap, and current stops everywhere in that loop at once. That's the entire principle behind a switch: it's just a controlled gap you can open or close.

Resistance — the restriction

Resistance is whatever in the circuit pushes back against that flow — again like a narrower section of pipe restricting how much water can get through even at the same pressure. Resistance is measured in ohms (Ω). A thick, short wire has very low resistance. A long, thin wire — or a component specifically built to resist current, called a resistor — has higher resistance.



A wide, short pipe resists flow far less than a long, narrow one.

Every real conductor has some resistance, even wires, though it's usually small enough to ignore in basic circuits. Resistors are components built specifically to add a known, controlled amount of

resistance where you need it — which turns out to matter constantly, because without resistance limiting it, current can rise to damaging levels almost instantly.

The relationship: Ohm's Law

These three quantities aren't independent — they're locked together by one of the most useful equations in all of electronics, Ohm's Law:

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V = I × R
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Rearranged, this also gives you:

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I = V / R
R = V / I
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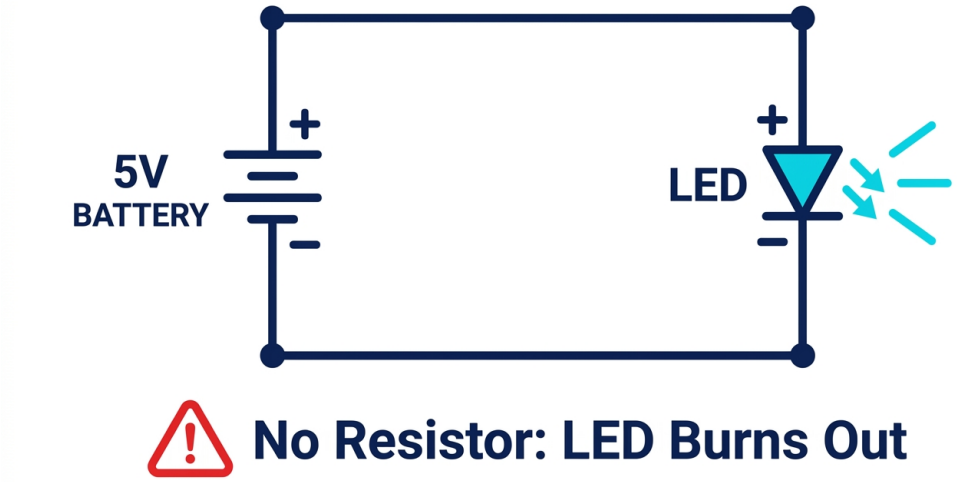


Ohm's Law, and how to rearrange it to solve for any one value.

Whenever you know any two of these three values, Ohm's Law gives you the third. This single relationship is the tool you'll reach for constantly — sizing a resistor, checking whether a component is safe to connect, or figuring out why a circuit isn't behaving the way you expected.

Putting it to work: sizing a resistor for an LED

Here's where this becomes genuinely practical. Say you're powering an LED from a 5V source. The LED itself needs about 2V across it to work correctly, and it's rated to safely handle 20 milliamps (0.02A) of current. Connect it directly to 5V with no resistance in the way, and it will draw far more current than it's rated for — and burn out almost instantly.



Connecting an LED directly to a voltage source with no resistor.

You need a resistor to absorb the difference. The voltage the resistor needs to handle is the leftover voltage: 5V source minus the 2V the LED uses, leaving 3V for the resistor. Using Ohm's Law, rearranged to solve for resistance:

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R = V / I
R = 3V / 0.02A
R = 150Ω
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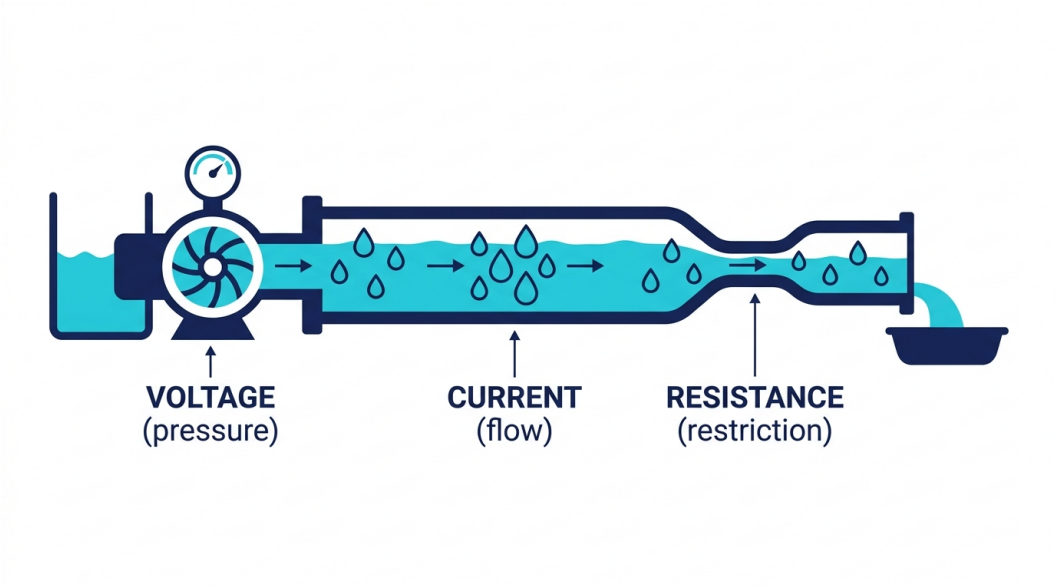


A correctly sized resistor splitting the voltage safely across the circuit.

A 150-ohm resistor placed in series with the LED limits the current to a safe level, protecting the LED while still letting it light up properly. This exact calculation — figuring out what resistor a component needs — is something you'll do constantly once you start building with real components.

Why this matters before you touch a sensor or motor

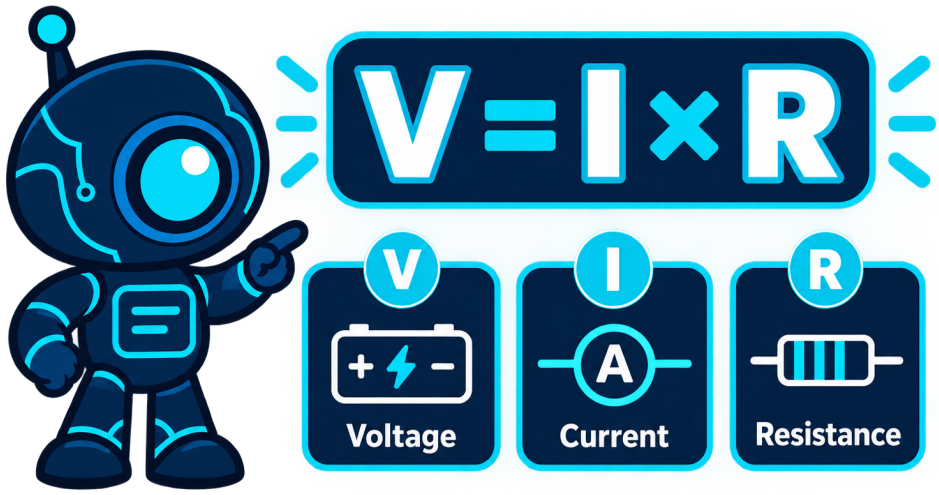
Every sensor and motor you'll work with later in this course has voltage and current requirements, and getting those wrong is one of the most common ways beginners damage components. Supply too much voltage, and you can destroy a sensor instantly. Draw too much current through a component not rated for it, and it overheats or fails. Ohm's Law is the tool that lets you check these numbers before you connect anything, rather than finding out the hard way.



Voltage, current, and resistance, all mapped onto one water-pipe system.

Where this leaves you

By the end of this lesson, you should understand what voltage, current, and resistance each represent, be able to picture them using the water-pipe analogy, and be able to apply Ohm's Law to solve for any one of the three when you know the other two — including sizing a basic resistor for a component like an LED. Next, you'll build on this by looking at how components get connected together: the difference between series and parallel circuits, and why that difference changes how voltage and current behave across a whole circuit.



Voltage, current, and resistance — tied together by Ohm's Law.